

4384_HES_Assignment_1

Blind Code: 4384

Paper Code: HES

Assignment: News and Views

Title: The digital aristocracy

Word count: 1494

The digital aristocracy

Our evolutionary instinct to follow leaders allows a small elite to gatekeep what society sees and shares online.

In the small-scale societies where human social learning evolved, copying a prestigious individual – like a skilled hunter or village elder – was a rational survival strategy requiring time and observation^{1,2}. But what happens when this ancient instinct collides with the frictionless architecture of the modern internet? A new Nature study by Niitsuma *et al.*³ shows prestige bias – our adaptive tendency to trust and imitate those perceived as successful – applies in online contexts much like in face-to-face societies³.

They show socially prestigious users on X (formerly Twitter) play a disproportionate and enduring role in shaping information diffusion, with a minority accounting for the majority of views and reposts. These findings suggest digital social networks amplify a familiar human bias for influential figures with profound implications for misinformation, public health messaging, and online platform design.

Until recently, work on prestige bias has focused largely on face-to-face interactions and relatively small-scale social networks – such as in the Solomon Islands². Yet, in this day and age, humans communicate and form networks across borders and time zones via social media⁴. Previous studies have spotlighted prestige bias online, showing users cluster around already-copied or expert-branded accounts, and that influencers successfully spread their own content; however, their power when diffusing others' content was left unclear^{5,6}.

Niitsuma and colleagues address this by investigating whether a proven ability to spread original content translates into the power to amplify others' work. Defining 'influencers'

as users with strong primary spread capabilities – quantified here by the ‘hg-index’, a metric capturing how consistently and explosively a user’s posts spread – the authors investigate whether individuals are statistically more likely to repost information received from high-status accounts than ordinary ones, or if such users simply appear powerful because they curate already popular content.

To test this, the authors analysed Japanese-language activity on X, reconstructing follower networks and tracking repost cascades over time to quantify secondary spread (the extent to which a repost generates further sharing). Comparing users across influence levels, they measured how far reposts travelled, how long they remained visible, and whether diffusion occurred through broadcast-like bursts or multi-step sharing chains.

Crucially, the authors addressed the sceptic’s natural objection: that famous people simply post better material. They used a statistical model controlling for posting time, follower counts, and differences in user activity and post type. This tested whether influencer status itself – rather than content or repost frequency – drives continued information spread.

The results strongly support the authors’ hypothesis that more influential users disproportionately drive secondary information spread: a tiny fraction of high-influence users generate over half of all views and reposts. Replicated in an English language dataset, these results seem to be robust. This influencer effect is not explained by more frequent reposts, but by reposts being far more likely to trigger further sharing. Importantly, increased repost engagement with more popular content indicates influence is greater when content has a strong appeal – prestige bias online may not be as uniform as it is in real life.

When content popularity was controlled, influential user reposts spread further and persisted longer than those shared by ordinary users. Importantly, influencers do not merely amplify content to their followers – they initiate deeper and more complex

cascades that keep information circulating across the network for longer, compared to ordinary users. This demonstrates the impact of prestige bias on information diffusion regardless of content type.

Statistical modelling also confirmed that influencer status remains a strong predictor of whether a repost will be shared again, after controlling for user and content differences. However, it found emotionally engaging or topical posts sometimes spread widely even when shared by low-influence users, indicating content driving some repost bias, as above. Regardless, no consistent content category matched the predictive power of user influence – prestige bias remains a dominant force shaping secondary spread. Notably, the same patterns of prestige bias were observed when the analysis was restricted to short time windows to limit algorithmic ranking effects, indicating influencer-driven amplification persists even in the absence of algorithmic promotion.

Figure 1 illustrates this bias schematically. In the left panel, a repost by an ordinary user (green node) reaches only a small number of immediate followers (yellow nodes), with little further propagation. In the right panel, a repost by an influential user triggers successive waves of sharing: larger circles indicate users with greater diffusion power, arrows represent reposting events over time, and colour gradients distinguish successive layers of secondary spread. This cascading structure highlights how influential users generate multi-step diffusion chains rather than simple one-step broadcasts – transforming passive audiences into active transmission chains.

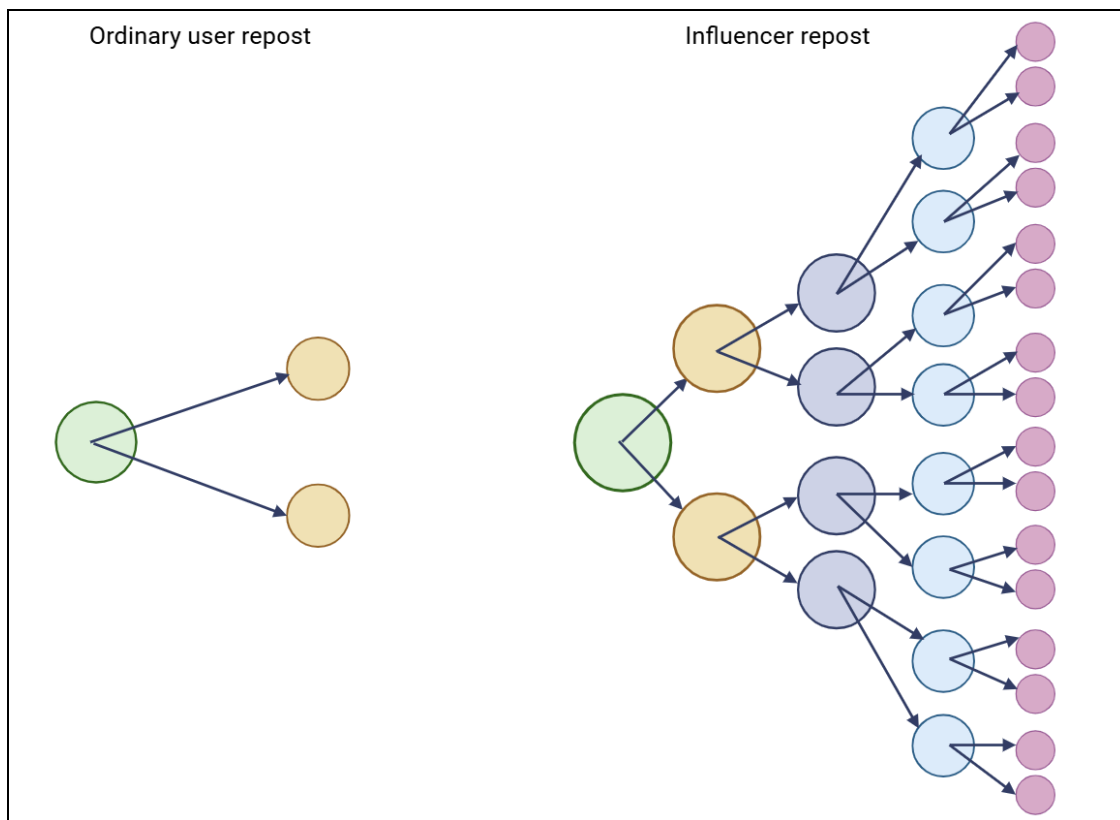


Figure 1: Diffusion patterns following reposts by ordinary and influential users. ‘Influencers’ spread information both more widely and persistently than ordinary users, whose reposts tend to fade after reaching immediate followers, creating shallow ‘broadcast’ patterns.

How confident can we be in these findings? Despite its scale and analytical sophistication, the study has several limitations. Influence is measured using an index derived from repost activity, raising the possibility of circularity and inflation of influence effects, as past visibility may partly predict future spread. Grouping half of all users into a single low-influence category may also mask meaningful variation in behaviour among ordinary users.

The reconstructed follower networks represent only a one-month snapshot, exclude private or official accounts, and rely on a chronological timeline that may not fully capture algorithmic curation – although a supplementary analysis partly accounted for this. The study also did not consider the role of bot activity in shaping repost cascades, which is important given its capacity to form clusters around influential users.

Fundamentally, the observational nature of the data means that causal claims should be interpreted cautiously, as experimental manipulation of exposure or influence was not possible. Nonetheless, whilst these constraints limit strong causal claims, they do not undermine the core conclusion that socially prestigious users exert disproportionate influence over how information travels online.

This study represents the first large-scale examination of prestige bias online whilst using innovative measures to track how reposts generate further sharing. The authors show how influential people of 'prestige' amplify content by simply reposting, not even creating it. Even further, a small elite of users drives roughly half of all secondary reposts, effectively acting as gatekeepers for what information spreads. By combining cultural evolution theory with computational social science, the paper demonstrates how ancient social learning strategies now operate on a global scale, revealing both vulnerabilities and responsibilities in modern information ecosystems.

Specifically, these findings unveil a profound evolutionary mismatch. Prestige bias evolved in small-scale societies as an adaptive mechanism for costly social learning, where prestige was slow to acquire and difficult to fake¹. Online platforms collapse these constraints - amplified prestige cues decouple influence from expertise, accelerating misinformation spread by influencers. This explains why dubious claims, memes, and trends can spread so rapidly online, and why corrective information can be far more effective when shared by influential users.

This has serious societal implications for public health messaging and democratic discourse, and is a pattern documented by similar studies showing that high-centrality users on other platforms, as well as official news and government accounts, can dominate retweet cascades^{7,8}. However, Niitsuma et al. reveal that this dominance is driven by transmission efficiency, not just audience size. This underlines the importance of understanding the role of influencers and algorithmic effects in information

networks, and is an opportunity for public health authorities to strategically leverage influential users to amplify accurate information and counter harmful narratives.

Further, the finding that content and post virality interact with user influence level indicates that, although a multiplier, the messenger is not a replacement for quality. This suggests the most effective strategy is to align engaging content with high-status spreaders, ensuring that the message is not just accurate, but carried by a voice the network trusts. Beyond leveraging prestige bias dynamics, the results of this paper challenge the structural foundations of social media platforms themselves. If algorithms reward sharing speed, they effectively amplify prestige bias. To fix this, platforms could hide follower counts or simply slow down the rate at which posts can be shared. This would stop influencers from triggering viral chains and give accurate information a fair chance to compete.

Future inquiries could broaden the scope of this study by testing prestige bias across different cultures and languages, whilst explicitly investigating the role of content quality. Methodological improvements are needed, including incorporating algorithmic timeline data, detecting bot activity, and applying causal inference techniques to strengthen claims about influence effects. Additionally, longitudinal data could reveal whether prestige bias persists beyond just one month's time, and exploring practical applications in misinformation correction, marketing, and public health would further extend the societal relevance of these findings. Nonetheless, the central message of the study is clear – human learning strategies once suited to intimate social settings now scale to global platforms, concentrating visibility and influence in the hands of a few.

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